

TESTBED SUBMISSION COVER SHEET**Phase 1 · Step 1.1 + 1.2 · Round R01**

Field	Entry
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Supervisor	Dr. Eric Opoku Osei
Submission round	R01
Date submitted	29th June 2026
Folder	ML
Model status	ENGINEERED
F/M/R/S operation	REPLACE — standard cross-entropy loss replaced with weighted cross-entropy loss
Dataset strategy	SINGLE-SOURCE
Claim tier ceiling	Tier 1 (Bounded)
Primary metric	Macro F1-score (classifier) · ROUGE-L + BERTScore F1 (RAG)
Statistical test	Wilcoxon signed-rank (paired, non-parametric, two-tailed, $\alpha=0.05$)
Runs × folds	20 runs × 1 stratified hold-out fold (70/15/15)
Random seeds	42, 123, 456, 789, 1024, 2, 7, 13, 17, 23, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71
Target journal (primary)	Computers & Education: Artificial Intelligence (Elsevier, Q1)
Format family	Elsevier
Passport attached	YES
Methods round	R01

Files Submitted

File name	Description
Agyei_Michael_Phase1_Method_R01.docx	Methods section Draft 0 — M1 to M21 per CAN-DO Lab template
Agyei_Michael_Phase1_Results_R01.docx	Results and Analysis Draft 0 — R1 to R10 per CAN-DO Lab template
Group1_Merged_Feedback_Reports.pdf	All three prior feedback reports merged: Architecture Audit + Dataset Advisory + Revised Assessment

Source Code and Reproducibility

GitHub repository: <https://github.com/maagyei04/Nyansapo-RAG-Research>

Notebook: Nyansapo_RAG_Pipeline_v2.ipynb (clean, reproducible, 17 cells)

Model weights: Zenodo DOI — pending upload prior to journal submission

Commit hash: To be inserted upon final push before submission

Negative Result Clearance Declaration

The following negative results are disclosed in R10 of the Results section, with diagnostic evidence per the CAN-DO Negative Results Playbook v2:

Finding: CRAG vs Vanilla LLM on ROUGE-L — $p=0.056$, not significant at Bonferroni-corrected $\alpha=0.0167$

Diagnosis (Cause 17): Power deficit. Achieved power = 0.446 at $n=200$. Required $n \geq 411$ for 80% power at $d=0.160$. Result labelled inconclusive, not 'no effect'.

Clearance conditions met: (1) Sign stable across evaluation corpus — CRAG > Vanilla on 102/200 queries. (2) Effect size $d=0.160$ positive throughout. (3) Power deficit diagnosed and stated. (4) Baseline received identical tuning budget. (5) ROUGE-L reported as primary RAG metric. (6) Test set scored once.

Finding: CRAG vs Naive RAG — $p=0.554$, not significant

Diagnosis: Effect size $d=0.026$, negligible. The CRAG evaluator did not produce a measurable improvement over naïve RAG at $n=200$ gold queries. Reported as a genuine boundary condition, not a failure.